FLAIR Task Schematic

AKI KDIGO Stage 2 within 24 Hours (Task T5)

FLAIR Benchmark

June 4, 2026

Abstract

We specify the FLAIR continuous AKI task: at every hourly decision point during the first ICU stay, predict whether the patient will meet the KDIGO stage 2 serum-creatinine criterion within the next 24 hours. The label, baseline rule, and cohort exclusions follow Koyner et al. (2018); urine-output criteria are not used. One cohort deviation (KDIGO/Koyner exclusion rule 2 — ICD CKD stage 4+) is documented because CLIF `Diagnoses` is not yet exposed via the FLAIR loaders.

1 FLAIR operational definition

1.1 Anchors and windows

Let H_i denote a hospitalization with first ICU stay window $[t_i^{\mathrm{ICU}_1,\mathrm{start}}, t_i^{\mathrm{ICU}_1,\mathrm{end}}]$ returned by `compute_icu_timing` from the CLIF `ADT` table.

Serum-creatinine baseline.
For each SCr measurement $\mathrm{SCr}_i(t)$ with $t =$ `lab_result_dttm`,

$$\mathrm{baseline}_i(t) \;=\; \begin{cases} \min\{\mathrm{SCr}_i(\tau) \,:\, \tau \in (t - 168\mathrm{h},\, t)\} & \text{if non-empty,} \\ \mathrm{SCr}_i(t_i^{\mathrm{index}}) & \text{otherwise (index SCr fallback)} \end{cases}$$

where t_i^{index} is the earliest in-stay SCr of H_i. The window is strictly left-exclusive of t (the current measurement is *not* part of its own baseline).

KDIGO stage 2 flag.
$\mathbb{1}_{\mathrm{KDIGO2}}(t) = \mathbb{1}[\mathrm{SCr}_i(t) \,\geq\, 2.0 \cdot \mathrm{baseline}_i(t)]$.

First-onset time.
$t^i_{\mathrm{KDIGO2}} = \min\{\, t \mid \mathbb{1}_{\mathrm{KDIGO2}}(t) = 1 \,\}$; $t^i_{\mathrm{KDIGO2}} = +\infty$ if the patient never meets KDIGO-2 during the first ICU stay.

Prediction-window start (pred_start).
$\mathrm{pred_start}_i = t_i^{\mathrm{ICU}_1,\mathrm{start}}$.

Prediction-window stop (pred_stop).
pred_stop$_i$ = min($t_i^{\mathrm{ICU}_1,\mathrm{end}}$, t^i_{KDIGO2}). The interval [pred_start$_i$, pred_stop$_i$) is closed-open; t^i_{KDIGO2} itself is *not* a decision point.

Decision-point grid.
Hourly grid: $D_i = \{\text{pred_start}_i + k\cdot 1\text{h} : k \in \mathbb{Z}_{\geq 0},\ \text{pred_start}_i + k\cdot 1\text{h} < \text{pred_stop}_i\}$.

Label (`label_aki_kdigo2_24h`).
For $d \in D_i$:

$$y_i(d) = \mathbb{1}\left[t^i_{\mathrm{KDIGO2}} \in (d,\ d + 24\text{h}]\right].$$

Patients who never meet KDIGO-2 ($t^i_{\mathrm{KDIGO2}} = +\infty$) contribute all-negative rows over [pred_start$_i$, $t_i^{\mathrm{ICU}_1,\mathrm{end}}$).

1.2 Cohort (Koyner 2018)

- Age at admission ≥ 18 with valid `admission_dttm` and `discharge_dttm`.
- ≥ 1 ICU ADT record (first ICU stay defined).
- ≥ 1 creatinine measurement during the first ICU stay (baseline must be computable).
- **Exclude** first SCr ≥ 3.0 mg/dL (Koyner rule 1; implemented).
- **Exclude** RRT within 48h of first SCr (Koyner rule 3; implemented when an RRT table is provided, skipped here as the RRT loader is not wired).
- **Not applied**: ICD CKD stage 4+ (Koyner rule 2). The CLIF `Diagnoses` table is not yet exposed via `flair_benchmark/_clif.py`; this is a documented deviation and is flagged in the main-paper Limitations.

1.3 CLIF tables and categories used

Task T5 reads four CLIF v2.1 tables. The cohort entry filter is on `adt.location_category`; the KDIGO-2 signal is computed from a single category-column filter in `labs`.

CLIF table	Category column	Literal values kept (case-insensitive)
`hospitalization`	—	cohort/demographics join; no category filter
`patient`	—	demographics join; no category filter
`adt`	`location_category`	{`"icu"`}
`labs`	`lab_category`	{`"creatinine"`} (loader-side filter)

Table 1: CLIF surface for Task T5. The `labs` filter is applied inside `load_labs(lab_categories=["creatinine"])` so non-SCr rows never enter memory.

Columns selected from each table:

- `hospitalization`: `hospitalization_id`, `patient_id`, `admission_dttm`, `discharge_dttm`, `age_at_admission`, `discharge_category`.

- `patient`: `patient_id`, `sex_category`, `race_category`, `ethnicity_category`, `death_dttm`.
- `adt`: `hospitalization_id`, `location_category`, `in_dttm`, `out_dttm`.
- `labs`: `hospitalization_id`, `lab_category`, `lab_result_dttm`, `lab_value_numeric`.

CLIF surfaces specified but not loaded (deviations). Two CLIF surfaces appear in the operational definition above but are not currently exercised by the build pipeline:

- **CLIF `patient_problem_list` (or Diagnoses).** Required to implement Koyner exclusion rule 2 (CKD stage 4+). Not exposed by `flair_benchmark/_clif.py`; the rule is skipped and the deviation is rendered as a gray-dashed node in Figure 1.
- **RRT signal source.** Koyner exclusion rule 3 expects an RRT-bearing table (a procedures-style table or `respiratory_support` with CRRT in `device_category`). `apply_koyner_exclusions` is currently called with `rrt_df=None`, so rule 3 is not enforced. Adding the loader is mechanical and would not change the schematic, only the realized cohort.

2 Algorithmic flow

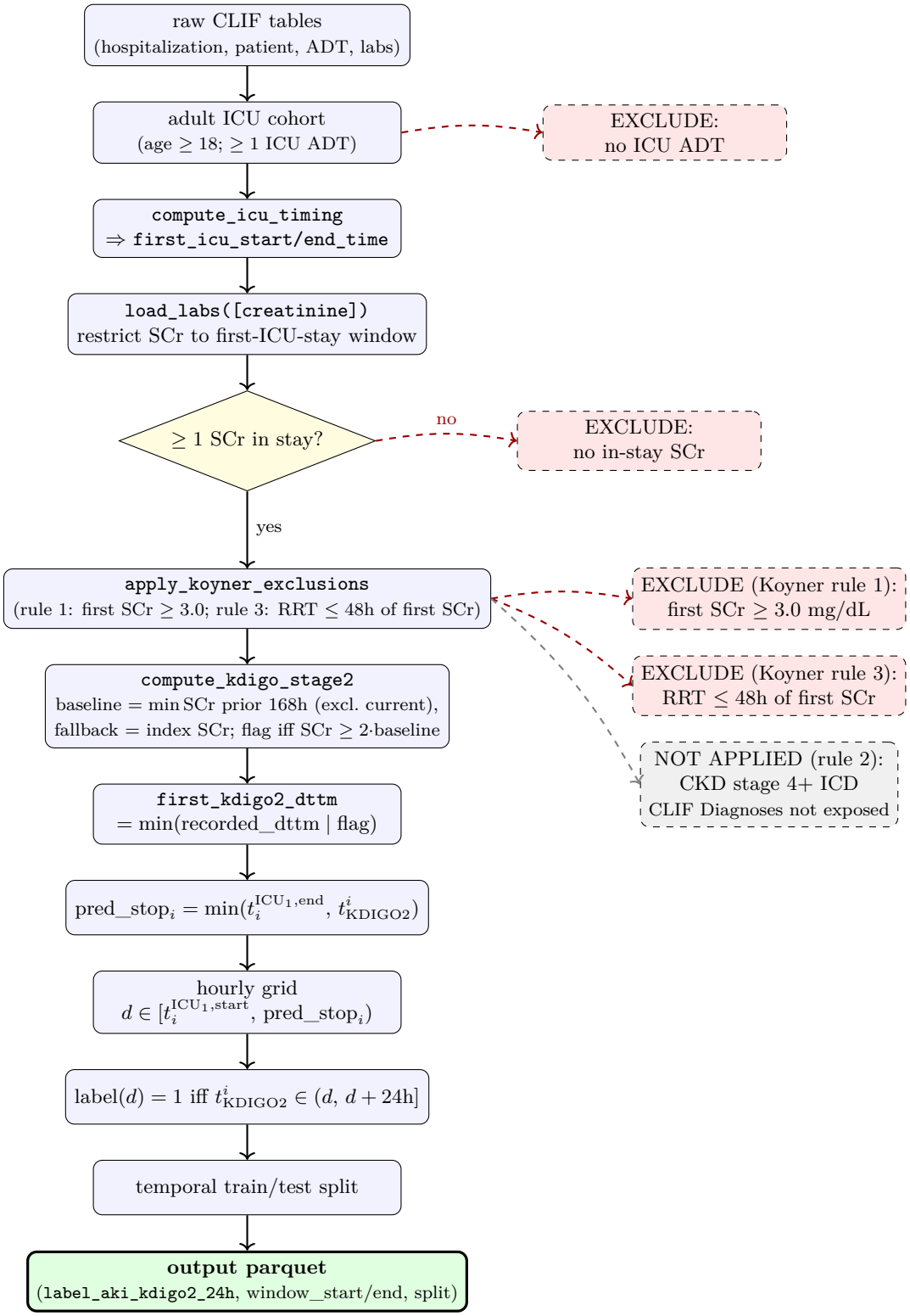


Figure 1: Build pipeline for Task T5. Solid arrows: main flow. Dashed red arrows: active exclusions. The gray dashed node marks the documented Koyner rule-2 deviation. The diamond gates hospitalizations on the existence of an in-stay creatinine measurement; without one the baseline (and therefore every KDIGO-2 flag) is undefined.

Step-by-step prose:

1. Load CLIF tables: `hospitalization`, `patient`, `adt`, `labs`.
2. Apply adult-ICU cohort filters (age, valid admission/discharge, ≥ 1 ICU ADT).
3. Compute first-ICU-stay window via `compute_icu_timing(adt)`.
4. Load `labs` restricted to `lab_categories=["creatinine"]`; project to `(hospitalization_id, recorded_dttm` $\leftarrow$ `lab_result_dttm, lab_value)`; drop nulls; restrict to the first-ICU-stay window.
5. Drop hospitalizations with no in-stay SCr (no baseline computable).
6. Apply `apply_koyner_exclusions(cohort, creat, rrt_df=None)`: rule 1 excludes first SCr $\geq$ `KOYNER_FIRST_SCR_EXCLUSION_MGDL` $= 3.0$; rule 3 is skipped because `rrt_df=None`.
7. Apply `compute_kdigo_stage2(creat, baseline_window_hours=168)`: rolling-min of prior 168h (exclusive of the current measurement via a 1-row shift), coalesce with the per-hospitalization index SCr, flag iff SCr $\geq$ `KDIGO_STAGE2_RATIO` $\cdot$ baseline $= 2.0 \cdot$ baseline.
8. Aggregate to $t^i_{\text{KDIGO2}} = \min(\text{recorded_dttm} \mid \text{flag})$ per hospitalization (null when no flag).
9. Compute $\text{pred_stop}_i = \min(t_i^{\text{ICU}_1,\text{end}}, t^i_{\text{KDIGO2}})$. Truncating at first onset eliminates post-onset decision points, which would otherwise be trivially positive.
10. Explode the hourly decision-point grid $[\text{pred_start}_i, \text{pred_stop}_i)$ per hospitalization (`pl.datetime_ranges`, closed-open).
11. Assign $y_i(d)$ per the label rule above.
12. Assign `window_start` $= d - 24$h, `window_end` $= d$ (fixed 24h lookback, matches the FLAIR convention).
13. Apply temporal train/test split per `flair_benchmark._split.assign_split`, keyed on `prediction_dttm`.
14. Select `OUTPUT_COLS` and persist as parquet.

3 Design choices and rationale

Why SCr-only (not urine output). KDIGO defines AKI by SCr *or* urine-output (UO) criteria, but hourly UO is recorded unevenly across CLIF sites (and across services within a site — floors generally do not record hourly UO at all). Including UO would force per-site, per-cohort cadence assumptions and make federated results incommensurable. Koyner et al. (2018) use an SCr-only criterion for the same reason; FLAIR follows. A UO-augmented sensitivity arm is out of scope for v1.

Why a 7-day rolling-min baseline with index-SCr fallback. The 7-day prior-min is the KDIGO baseline rule operationalized for inpatients without a pre-admission SCr record. The index-SCr fallback catches admissions whose first SCr is the only measurement in window and is the convention in Koyner et al. (2018) and Tomašev et al. (2019). The current row is excluded from its own baseline window via a 1-row shift within hospitalization (see `_aki.py`); without that shift, a single high-SCr reading could anchor its own baseline and silently suppress the flag.

Why truncate predictions at $t_{\mathbf{KDIGO2}}$. After onset, any prediction is trivially correct (the event has already occurred). Unlike the sepsis task (where the truncation also removes a label-leakage channel from a clinician action), the AKI truncation is purely about not double-counting an event after the fact: KDIGO-2 is a lab-derived state, not a clinician action, so post-onset decision points have no leakage risk — they are simply uninformative.

Why the CKD rule-2 deviation is acceptable for v1. Rule 2 excludes patients whose AKI is masked by chronic kidney disease stage 4+. Without ICD codes from the CLIF `Diagnoses` table (not yet exposed via `flair_benchmark/_clif.py`), we cannot apply the rule. The first-SCr-≥ 3.0 filter (rule 1) catches the majority of stage 4+ patients in practice via their baseline creatinine, but a small false-positive tail remains. The deviation is flagged in the schematic flowchart (gray dashed node) and in the main paper's Limitations.

Prevalence implications. Hourly grids over an ICU stay produce highly imbalanced labels: MIMIC-CLIF yields a per-row positive rate of 1.7% (65,577 stays / 4,135,079 hourly rows; 4.8% of stays ever positive — see `docs/paper/outline.tex` Table 1). Reporting AUPRC alongside AUROC is mandatory, per the `META.evaluation_metrics` convention.

4 Out of scope

- KDIGO stages 1 and 3 labels.
- Urine-output AKI criterion.
- Hospital-wide (non-ICU) AKI prediction.
- Multi-onset / recurrent-AKI labels within a stay.
- Implementation of the CKD-stage exclusion (blocked on CLIF `Diagnoses` loader).
- Input feature design.

References

Jay L. Koyner, Kyle A. Carey, Dana P. Edelson, and Matthew M. Churpek. The development of a machine learning inpatient acute kidney injury prediction model. *Critical Care Medicine*, 46(7): 1070–1077, 2018. doi: 10.1097/CCM.0000000000003123.

Nenad Tomašev, Xavier Glorot, Jack W. Rae, Michal Zielinski, Harry Askham, Andre Saraiva, Anne Mottram, Clemens Meyer, Suman Ravuri, Ivan Protsyuk, et al. A clinically applicable approach to continuous prediction of future acute kidney injury. *Nature*, 572:116–119, 2019. doi: 10.1038/s41586-019-1390-1.